

2017 Indicators of the U.S. Biobased Economy

We have answered the questions and made the appropriate changes. We have transcribed each comment in bold characters to give and appropriate answer in normal characters. Page numbers refer to the revised version of the report.

Requested changes:

General comments

- 1. Several reviewers noted that the first chapter is called “Agricultural Indicators” but does not cover many areas of agricultural production, such as fruits, vegetables, livestock, and dairy. Rename the first chapter, “Field Crop Indicators.” Also clarify that we are leaving out many specialty crops.**

Response: The first chapter has been renamed as “Field Crop Indicators” and the following sentence has been added in page 1: “This report analyzes field crops and leaves out specialty crops because it is focused on those agricultural products from which bioproducts can produce (i.e., biofuels from corn).”.

- 2. The report uses the NASS statistic “Land in Farms” but renames it “Acreage in Agriculture.” “Land in Farms” includes not just cropland, and pastureland, but also includes woods, ponds, roads, buildings, etc. The NASS statistic “Cropland” will work better in the bioindicators report since we want to compare the acreage of particular field crops to a larger whole. The term “cropland” should also be used in the report.**

Response: The term “acreage in agriculture” has been removed from the report and the statistic “Cropland” has been added in figure 1.1.3 and table 1.1.1.

- 3. The report conflates field crop production with agricultural activities. We need to clarify that cropland is concentrated in the Midwest, Plains, and Delta. Other agricultural activities, such as animal feeding, dairy, and specialty crops have significant presence outside the Midwest.**

Response: The report has been modified to specify that we are analyzing cropland and not agricultural activities. Figure 1.1.3 and table 1.1.1 have been modified to show just cropland and not land in farms.

- 4. Update values from 2012 Census of Agriculture values from 2017 Census of Agriculture where applicable.**

Response: Figures 1.1.2, 1.2.11, 1.3.9, 1.4.9, 1.5.9, 1.6.10, 1.7.9 and tables 1.2.3, 1.3.3, 1.4.3, 1.5.3, 1.6.3, 1.7.3 have been updated with 2017 data.

Specific comments

p. 1: This discussion focuses entirely on field crops while ignoring fruits, vegetables, potatoes, sugarcane, sugarbeets, and mushrooms. Some also use significant amounts of water and pesticides. The report either needs a note discussing its crop coverage, or it needs to include important specialty crops.

The following sentence has been added in page 1: "This report analyzes field crops and leaves out specialty crops because it is focused on those agricultural products from which bioproducts can produce (i.e., biofuels from corn).".

p. 1, 3rd bullet, under "Agriculture": Make sure to note the geographic disparity (i.e., west versus east) in the share of freshwater used for irrigation. The differences are huge east and west of the Mississippi.

Response: The following sentence has been added: "81 percent of the total irrigation withdrawals and 74 percent of irrigated acres were in the western states, which are states with low average annual precipitation and need supplemental water to support crops."

p. 1, 3rd paragraph: Change 'The Chief and Economist' to 'The Chief Economist'.

Response: the manuscript has been changed appropriately.

p. 1: On a crop-year basis in 2017, 37.9 percent of total corn use was for ethanol. It is better to compare corn used for ethanol to total use rather than total production. Using production ignores corn available from stocks and imports that could be used for ethanol production.

Response: The corn used for ethanol production has been compared to total use. The sentence has been modified as follows "Nearly half of the corn used in the U.S. was used for ethanol production, and 27 percent of corn oil was used for biodiesel production."

p. 24: The term "acreage of agriculture" is misleading. These figures reflect land in farms, including homesteads, other buildings, woods, roads, etc. Since the report is subsetting major field crop acreage, it is better to use cropland.

Response: The summary table has been changed from "Acreage of agriculture" to "Cropland".

p. 24: The summary table of crop values only includes field crops. There are 14 specialty crops with \$1 billion or more of value, and 6 of those are valued at more than \$3 billion. Furthermore, California, where field crops are a minor part of the total, leads the state panel on the right.

Response: A note has been added in the executive summary that specifies that only field crops are analyzed. Furthermore, the summary table has been updated and only shows statistics of field crops.

p. 24: Note more prominently that these are only annual changes. The executive summary and remainder of the report are based on a broader timeframe.

Response: The summary table has been modified to specify that these are annual changes.

p. 24: Clarify whether the export value of \$135 billion includes specialty crops. Rank crops by value, not by tonnage.

Response: the summary table has been updated and only shows statistics of field crops and the crops have been ranked by value.

p. 25: The white arrows showing percent change in the bottom half of page 25 are actually the percent change of a percent change. This is confusing for the reader. Eliminate the percent change next to the white arrows, and just show the direction of change in this case.

Response: The summary table has been changed appropriately.

p. 27: Figure 1.1.2: “The bars represent the total acreage used for crops, pasture, and grazing in the United States from 2008 to 2017 (in million acres; left axis), and the line the average farm size in the United States from 2008 to 2017 (in acres; right axis) (1, 2).” The chart actually shows land in farms; about 80 million of those acres are in woodland not used for pasture and farmsteads, buildings, ponds, roads, wasteland. The circle chart is correct, but the figure title is not.

Response: The figure title has been modified as follows: “The bars represent the total land in farms in the United States from 2008 to 2017 (in million acres; left axis), and the line the average farm size in the United States from 2008 to 2017 (in acres; right axis) (1-3).”.

p. 27: “Agricultural activities are focused primarily in the Midwest region of the United States (Figures 1.1.3 and 1.1.4). The top 5 states by acreage are Texas, Montana, Kansas, Nebraska, and New Mexico (see Table 1.1.1 for more detail). The 5 states with more farms are Texas, Missouri, Iowa, Oklahoma, and California (see Table 1.1.2 for more detail).” This statement is misleading. Cropland is focused primarily in the Midwest, Plains, and Delta. Many high-value agricultural activities, like specialty crops, poultry, and dairy, are located in the West and East.

Response: Figure 1.1.3 and table 1.1.1 in page 28 have been modified to show cropland, and the sentence has been modified as follows: “Cropland and total farms are focused primarily in the Midwest region of the United States (Figures 1.1.3 and 1.1.4). The top 5 states with highest cropland are Texas, Kansas, North Dakota, Iowa, and Illinois (see Table 1.1.1 for more detail). The 5 states with most farms are Texas, Missouri, Iowa, Oklahoma, and California (see Table 1.1.2 for more detail).”.

p. 27, Figure 1.1.2: Change ‘The 5 states with more farms..’ to ‘The 5 states with most farms..’

Response: the manuscript has been changed appropriately.

p. 29, Figure 1.1.5: Change ‘..a brief declined..’ to ‘..a brief decline..’

Response: the manuscript has been changed appropriately.

p. 29, Figure 1.1.6: ‘.. when it reached its maximum of 150 billion dollars.’ That reads as if 150 billion dollars was a cap or ceiling. Also, change ‘..with highest.’ to ‘..with the highest..’

Response: The sentence has been modified as follows: “The value of exports increased until 2014 when it reached a maximum of 150 billion dollars. Exports dropped to 133 billion dollars in 2015 but recovered the positive trend with exports of 138 billion dollars in 2017. Canada is the country with the highest trade of agriculture goods with the United States, followed closely by China and Mexico (Figure 1.1.6).”

p. 30: “Although the total agriculture acreage has remained constant over the years, the net farm income in the United States has dropped over 40 percent, from 114 billion dollars in 2011 to 64 billion dollars, in 2017 (Figure 1.1.8).” The period 2011-13 was an historic peak for NFI; this should be noted.

Response: The sentence has been modified as follows: “Although the total agriculture acreage has remained constant over the years, the net farm income in the United States has dropped over 40 percent, from 2011 to 2017. In 2013, the net farm income reaches an historic peak at 124 billion dollars compared with 64 billion dollars in 2017 (Figure 1.1.8).”

p. 30, Figure 1.1.7: Change ‘..imports of agriculture in the United States..’ to ‘..imports of agricultural products in the United States..’

Response: the manuscript has been changed appropriately.

p. 30, Figure 1.1.9: Change ‘..income of farms operation is..’ to ‘..income of farm operations is..’

Response: the manuscript has been changed appropriately.

p. 32, heading: Change ‘Environmental’ to ‘Environmental indicators.’

Response: the manuscript has been changed appropriately.

p. 32: Consider changing ‘..accounting for approximately 80 percent..’ to ‘..historically accounting for approximately 80..’. This estimate, obtained from an ERS web source (author since retired), was originally derived from a USGS estimate of consumptive water use by sector for 1995. USGS continues to report water withdrawals by sector, but has since discontinued estimates of consumptive use across all U.S. sectors. (The ERS web source text will be revised with the next update.)

Response: The sentence has been modified as follows: “The agriculture sector is a major user of both ground and surface water in the United States, historically accounting for approximately 80 percent of the country’s consumptive water use (9).”

p. 32, final paragraph: The two acreage estimates cited represent two sources, but also two separate years, 2012 and 2015. Be clear about the different methodologies and production years.

Response: The text was updated to include a notation of the different production years (2012 and 2015) and methodologies/production agencies (i.e., USDA, U.S. Geological Survey) as follows: “The most recent Census of Agriculture, conducted in 2012, reported total U.S irrigated acreage at 55.8 million acres (3), while a more recent study conducted in 2015 reported the

same metric at 63.5 million acres (10). Though the two studies were conducted in different years by different agencies, both documents are concurrent in noting that five states – Arkansas, California, Idaho, Nebraska, and Texas – account for over half of the nation’s irrigated agriculture acreage.”.

p. 36: The opening sentence seems to apply broadly to both chemical and organic (eg. manure) fertilizers, while the graphic seems to cover only chemical fertilizers. Clarify whether the quantities in the chart include organic fertilizers.

Response: The figure caption was updated to specify that the chart depicts total chemical fertilizer use as follows: “Total chemical fertilizers applied to farms in the United States from 2008 to 2014 (in million metric tons) (14).”

Furthermore, the opening sentence on page 36 was edited for clarity, given the reviewer’s remarks.

p. 36: NASS Agricultural Chemical Use Program relies on sample of farmers. ERS data relies on the entire population of farmer hence the coverage is different. The discussion in page 36 does not make this clear and it might lead to confusion.

Response: The text was updated to clearly denote the differences between the NASS and ERS data, as outlined by the reviewer. Furthermore, Figure 1.1.16 clearly denotes which data source is utilized for the graphic.

p. 36, last paragraph: Another factor that led to a decrease in fertilizer consumption was the financial crisis of 2008-2009. Reduction in crop exports and increased financial constraints affected temporally crop production which in turn decreased fertilizer consumption during the period.

Response: This is a valid point by the reviewer, and the text was updated to include the impact of the financial crisis. An additional citation was also included to further support this point. The paragraph has been updated as follows: “The amount of fertilizer applied to farms has remained nearly constant over the years, with no persistent trend save for a dip in total fertilizer consumption in 2009 (Figure 1.1.16). The decline in fertilizer consumption in 2009 is attributable to both record fertilizer prices in 2008 and impacts from the 2008-2009 economic crisis, which resulted in a weakening of global demand for U.S. agriculture (14, 15). Lower demand for crops, coupled with record fertilizer prices, led farmers to reduce their fertilizer usage the following year (see Figure 1.1.17).”.

p. 37: ‘Matching application rates to crop needs is an important component of optimizing crop production’. This is also important for reducing nutrient emissions and loadings to water bodies—a point that seems central to a section on environmental indicators.

Response: The text was updated to include a variation of the reviewer’s comment; that is, optimizing application rates not only helps plant production but also reduces nutrient loading. The following sentence has been added: “Matching application rates to crop needs is an important component of optimizing crop production, in addition to reducing nutrient exports and loadings to nearby waterbodies.”.

p. 40: Again, compare corn use for ethanol to total corn use, not production.

Response: We have compared corn used for ethanol to total corn use. The sentence has been modified as follows: "Almost half of the corn consumed in 2017 was used for ethanol production, and 27 percent of corn oil was used for biodiesel production, which makes corn the most relevant crop in terms of biofuel production."

p. 40, 4th paragraph: "...downward trend in relative corn prices" Relative to what? Prices declined in absolute value.

Response: The sentence has been modified as follows: "Although the production of corn has remained nearly constant over the years, earnings from corn have been reduced in recent years due to the downward trend in corn prices."

p. 41: It's misleading to say *corn acreage devoted to ethanol*, the use of corn is generally not determined as it is planted, but as it is marketed. While some corn is grown under production contracts (popcorn and seed corn for example) the disposition of most corn is not determined when it is planted.

Response: The sentence has been removed and Figure 1.2.1 has been modified appropriately.

p. 43: Corn is mostly used for feed, ethanol, and exports.

Response: The following sentences has been added: "Most of the corn grown in the U.S. is used for feed, fuel alcohol and exports. In particular, 15 percent of corn produced in the U.S. is being exported, 37 percent of corn is used for ethanol production and 37 percent for animal feed in 2017."

p. 44: The statement at the top of the page neglects the effects of yield on crop production. Especially during the aforementioned drought years, this can have a large impact on the relationship between acreage and production. Also, this statement ignores corn which is harvested for silage. According to the NASS Grains Crushing and Co-Products Production report, total corn oil production in 2017/18 was a little over 3 million.

Response: The sentence has been modified as follows: "Most corn production, 87 percent of the total, is produced in the Midwest. Iowa, Illinois, Nebraska, Minnesota, and Indiana are the top 5 states for corn production (Figure 1.2.5 and Table 1.2.1). This is consistent with the geographical information about corn acreage."

p. 46: Note that much of the imported corn is either organic or seed corn.

Response: The following sentence has been added: "Most of the corn imported in the U.S. is organic corn and seed corn."

p. 49, Figure 1.2.13: This chart shows the value of corn oil, but reference 10 is for the price of corn, not the price of corn oil.

Response: The reference has been updated to "United States Department of Agriculture - USDA - Economic Research Service Oil Crops Outlook: January 2018. Available at: <https://www.ers.usda.gov/publications/pub-details/?pubid=86740> [Accessed April 1, 2018]."

p. 67: The sorghum section needs to mention that sorghum is used for fuel ethanol production.

Response: The sentence has been modified as follows: "Sorghum is a cereal plant of the grass family and it is the third largest grain crop in the United States after corn and wheat. Sorghum, which is a gluten free grain, is commonly used for human consumption (i.e., flatbreads, cakes, sweetener), as animal fodder, to produce alcoholic beverages and to produce fuel ethanol (1)."

p. 122, figure 2.2.5: The export country maps show regions, some of which are outlined in thick white while Brazil regions are outlined in a thin white line. The thin white line is preferable, and the maps should be uniform. This applies to other export country maps as well.

Response: the figures have been changed appropriately.

p. 128, table 2.3.1: "Infraestructure" should be "Infrastructure"

Response: The table has been changed appropriately.

p. 135: "Just one third of total U.S production of wood pellets is consumed domestically, and it is used primarily for heating." From Figure 2.4.4. this appears to be true for 2016, but not for 2017 when domestic consumption is much less than one-third. It should be clarified that the domestically consumed fraction will vary with winter temperature and growth in exports.

Response: That is right, the one-third ratio is an average measure, the sentence has been changed has follows: "Just 20 percent of total U.S production of wood pellets is consumed domestically, and it is used primarily for heating. In particular, wood pellet domestic consumption accounted for 148 trillion BTUs for electricity generation and 1,997 trillion BTUs for heating in 2017 (10). Note that the domestically consumed ratio varies with winter temperatures and the growth in exports."

p.141: "...which is the Britain's larger power station" should be rephrased to "which is one of Britain's larger power stations."

Response: the manuscript has been changed appropriately.

p. 144: "Before burned in a waste-to-energy plant" should be "Before being burned" or "Before burning"

Response: The sentence has been modified as follows: "Before being burned in a waste-to-energy plant, hazardous and recyclable materials are separated from municipal solid waste (1)."

p. 194: Delete the mention of ethanol infrastructure. REAP is a renewable energy and energy efficiency program. Singling out ethanol infrastructure makes it sound like REAP is an ethanol program.

Response: The sentence has been modified as follows: "The Rural Energy for America Program (REAP) provides grants and loan guarantees to agricultural producers and rural small businesses to purchase renewable energy systems or make energy efficiency improvements (7)."

p. 194: Grant maximums are \$500,000 for renewable energy and \$250,000 for energy efficiency. Grants are competitive, awards are not certain.

Response: The sentence has been modified as follows: “The maximum loan guarantee is 25 million dollars, and the maximum grant funding is 500,000 dollars for renewable energy and 250,000 dollars for energy efficiency.”

p. 195: Note that the Repowering Assistance Program was eliminated in the 2018 Farm Bill.

Response: The following sentence has been added: “The Repowering Assistance program was eliminated in the 2018 Farm Bill.”

p. 197: The Biorefinery Assistance Program has never provided grants, only loan guarantees.

Response: The sentence has been modified as follows: “The Biorefinery Assistance Program provides loan guarantees for the development, construction, and retrofitting of commercial-scale biorefineries that produce advanced biofuels (26). Eligible applicants include individuals, state or local governments, farm cooperatives, national laboratories, institutions of higher education, and rural electric cooperatives. The maximum loan guarantee is 250 million dollars (33). The loan guarantees are available through the end of fiscal year 2019.”

p. 197: The Biodiesel Education Program was not funded for FY2019; funds were available through FY2018.

Response: The following sentence has been added: “The biodiesel education program was available through fiscal year 2018.”